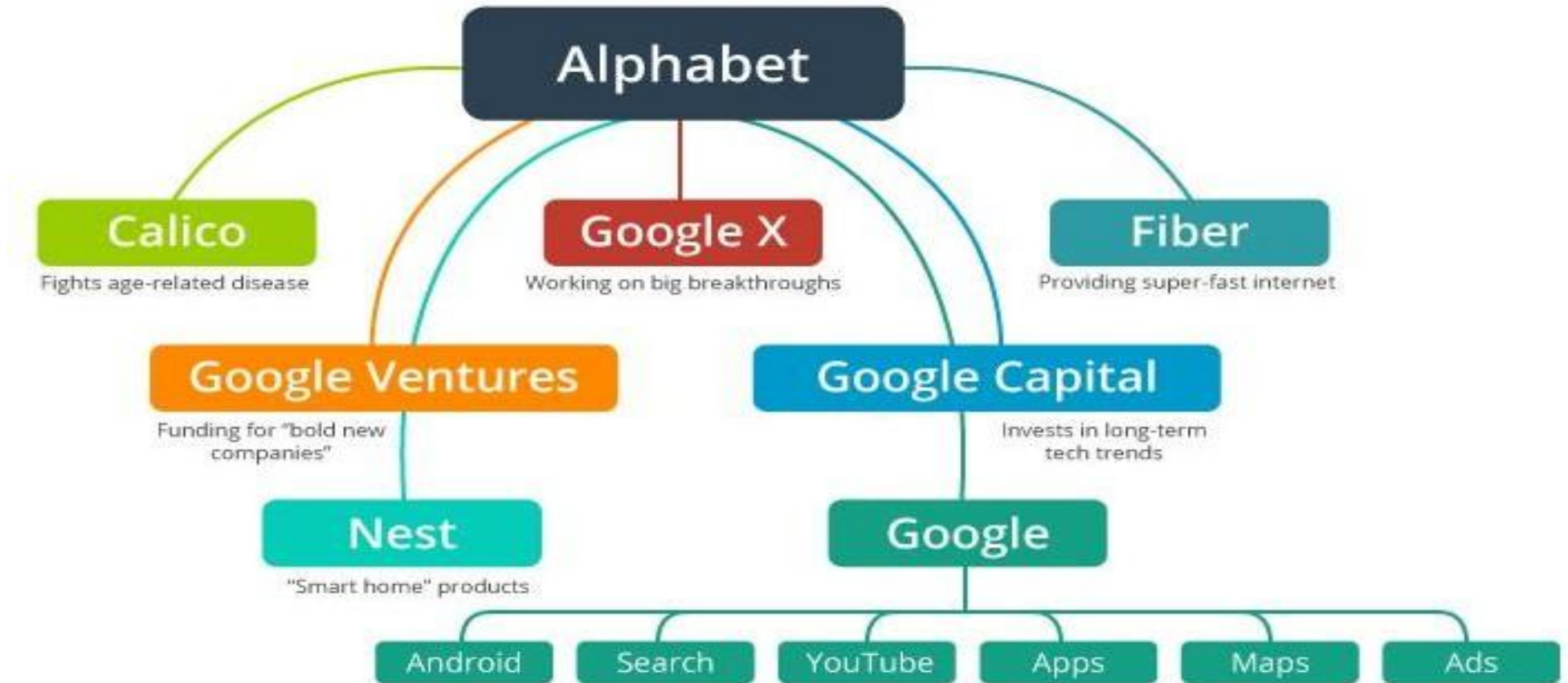


Elements of Project Management



Detailed Syllabus		
Unit	Description	Duration
1.	Introduction to Project Management Definition, classification of project management in different environments, scope, systems approach, project feasibility, project life cycle, systems development cycle phases.	08
2.	Project Organization Project organization, types of structures, roles of project managers and team, integration in large scale projects.	05
3.	Project Planning and Scheduling Planning fundamentals, project master plan, work breakdown structure and other tools of project planning, work packages, project organization structure and responsibilities, responsibility matrix, use of Gantt charts, critical path method (CPM) and project evaluation and review technique (PERT) for scheduling and resource allocation, introduction to software tools for documentation, scheduling, analysis etc.	12
4.	Cost Estimating and Budgeting Cost estimating process, elements of budgeting, project cost accounting and management information systems, cost schedules and forecasts.	06
5.	Project Risk Management Risk concepts and identification, risk response planning, risk management methods, information monitoring, control process, cost accounting systems for project control, performance analysis.	09
6.	Evaluation and Termination Project evaluation and reporting, closing the contract.	05
	Total	45

Text Books

1. Jack R Meredith, Samuel J Mantel, Shafer *Project Management –A Managerial Approach*, 8th Edition, Wiley Publications, 2017.
2. Erik W Larson, Clifford F Gray, Rohit Joshi, *Project Management the Managerial Process*, 8th Edition, McGraw Hill, 2020.

Reference Books

1. Dennis Lock, *The Essentials of Project and Programme Management*, 4th Edition, Routledge Publications, 2017.
2. Prasanna Chandra, *Projects: Planning, Analysis, Selection, Financing Implementation and Review*, 8th Edition, McGraw Hill Education, 2017.

Additional Reference Book:

J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology-Principle & Practice*, 5th Edition, Routledge (Taylor & Francis), 2019

NPTEL online course videos

1. Prof. Ramesh Anbanandam – IIT Roorkee
https://www.youtube.com/playlist?list=PLLy_2iUCG87CBuNhvti0h6W54ZmqrSDMJ
2. Prof. R.N. Sengupta -IIT Kanpur
<https://www.youtube.com/playlist?list=PLBd76GK9sWTwVXm9FIVHOTXXbGY2vZR8z>
https://www.youtube.com/watch?v=PqQqTAu_FiM&t=5952s

Note: Project management knowledge draws on nine areas, viz., integration, scope, time, cost, quality, procurement, human resources, communications and risk management.

- ❖ Organizations are systems of human and physical elements created to achieve goals
- ❖ *An **organization** can be defined as a group of individuals who coordinate their activities in order to accomplish a business and/or social objective*
- ❖ **Organizational structure** is defined as the framework of roles, responsibilities, authority relationships, and communication channels within an organization
- ❖ ***Organization planning** is a process of identifying, documenting, and assigning project roles, responsibilities and reporting relationships*
- ❖ Selecting/Choosing an appropriate organizational structure facilitates the effective implementation of an organizational plan
 - Concepts of organizational structure apply to all kinds of organizations (companies, institutions, agencies) as well as to their subunits (divisions, departments, projects, and teams)

❖ The structure of an organization depends on parameters-

- Rate of change of technology
- Availability of resources
- The product/service sold
- Competition
- Decision making requirements
- Size of the Organization
- Environment (legal, political, social and economic)
- Culture (refers to the shared values, beliefs, customs, and practices that determine the behavior of members within a particular organization)

●● **Authority** is the power granted to individuals (possibly by their position) so that they can make final decisions.

●● **Responsibility** is the obligation incurred by individuals in their roles in the formal organization to effectively perform assignments.

●● **Accountability** is being answerable for the satisfactory completion of a specific assignment. (Accountability = authority + responsibility)

Organization Structure:

- Represents the management hierarchy:
 - Reporting relationships who report to whom
 - The official chain of control or authority, which deals of official activities, such firing, hiring, and promotion.
- Organization can be grouped into major subdivisions on the basis of a number of frameworks.
- Responsibilities & type of work for each subdivision
- Official lines of authority & communication

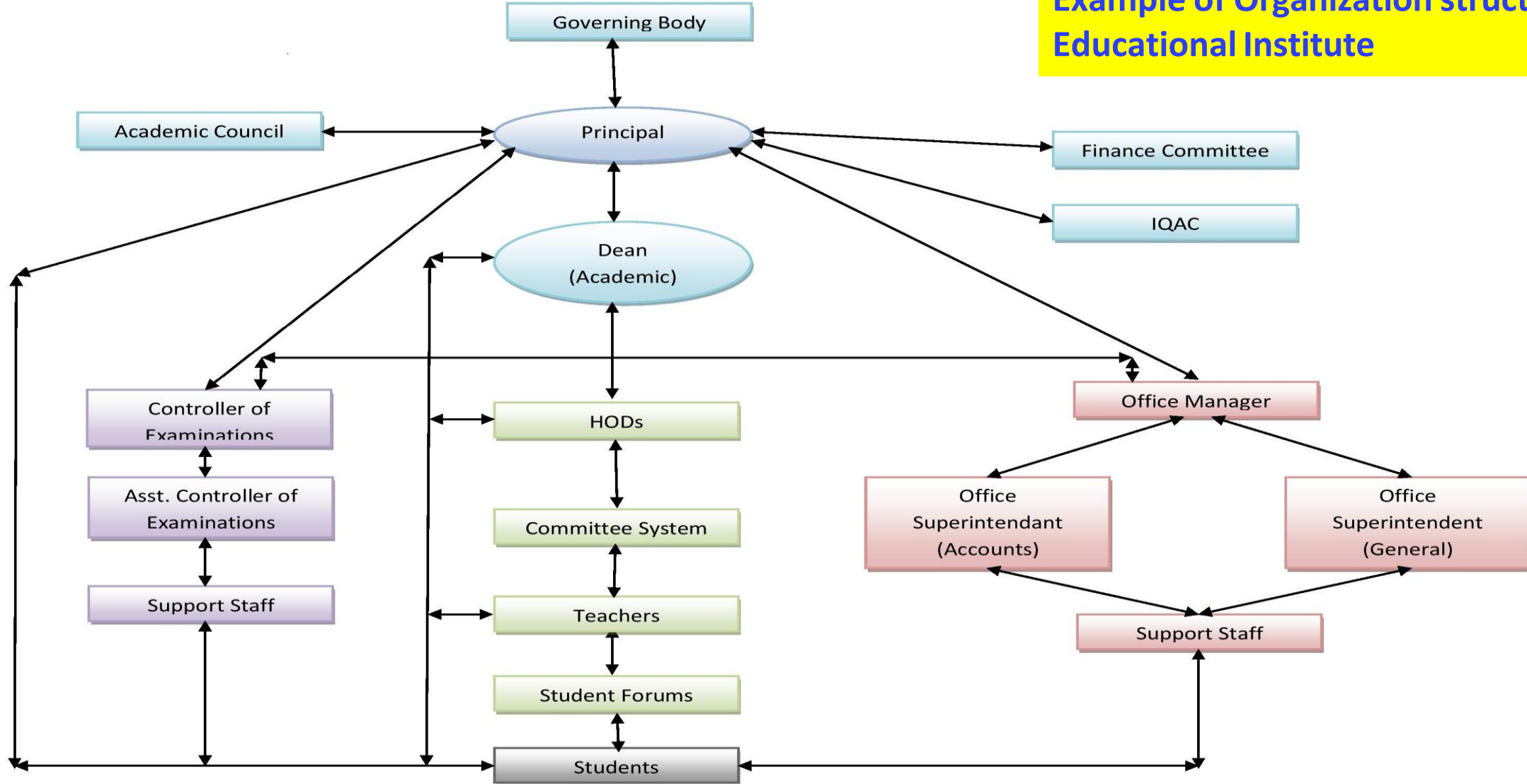
Informal organization also important and can exist in parallel to the official organization, communication of the informal is socially motivated.

Different Organizational Structures*:

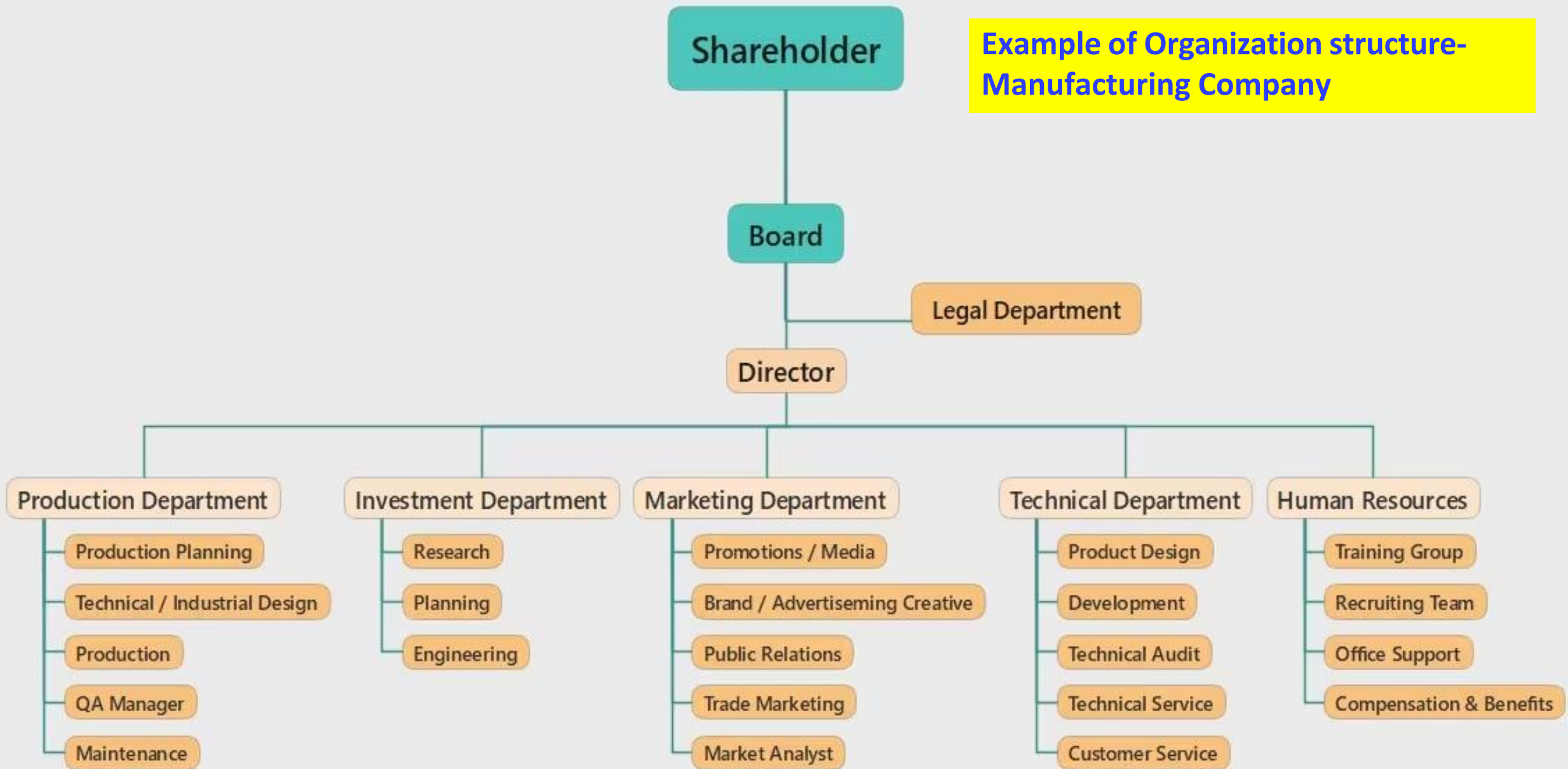
- Traditional (classical) organization (also refer as functional organization)
- Pure Product (projectized) organization
- Matrix organization

Internal Organization Structure

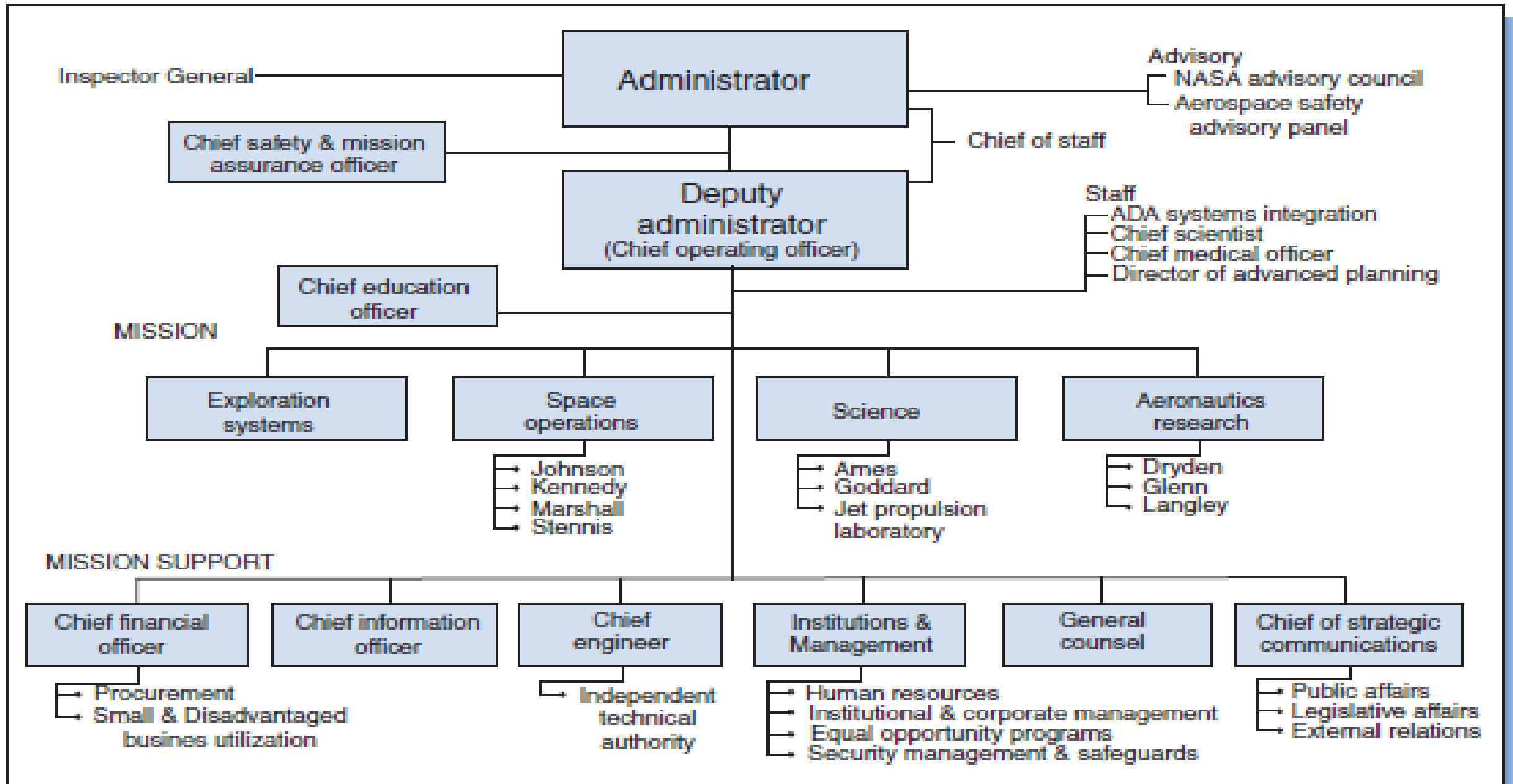
Example of Organization structure- Educational Institute



Example of Organization structure-
Manufacturing Company

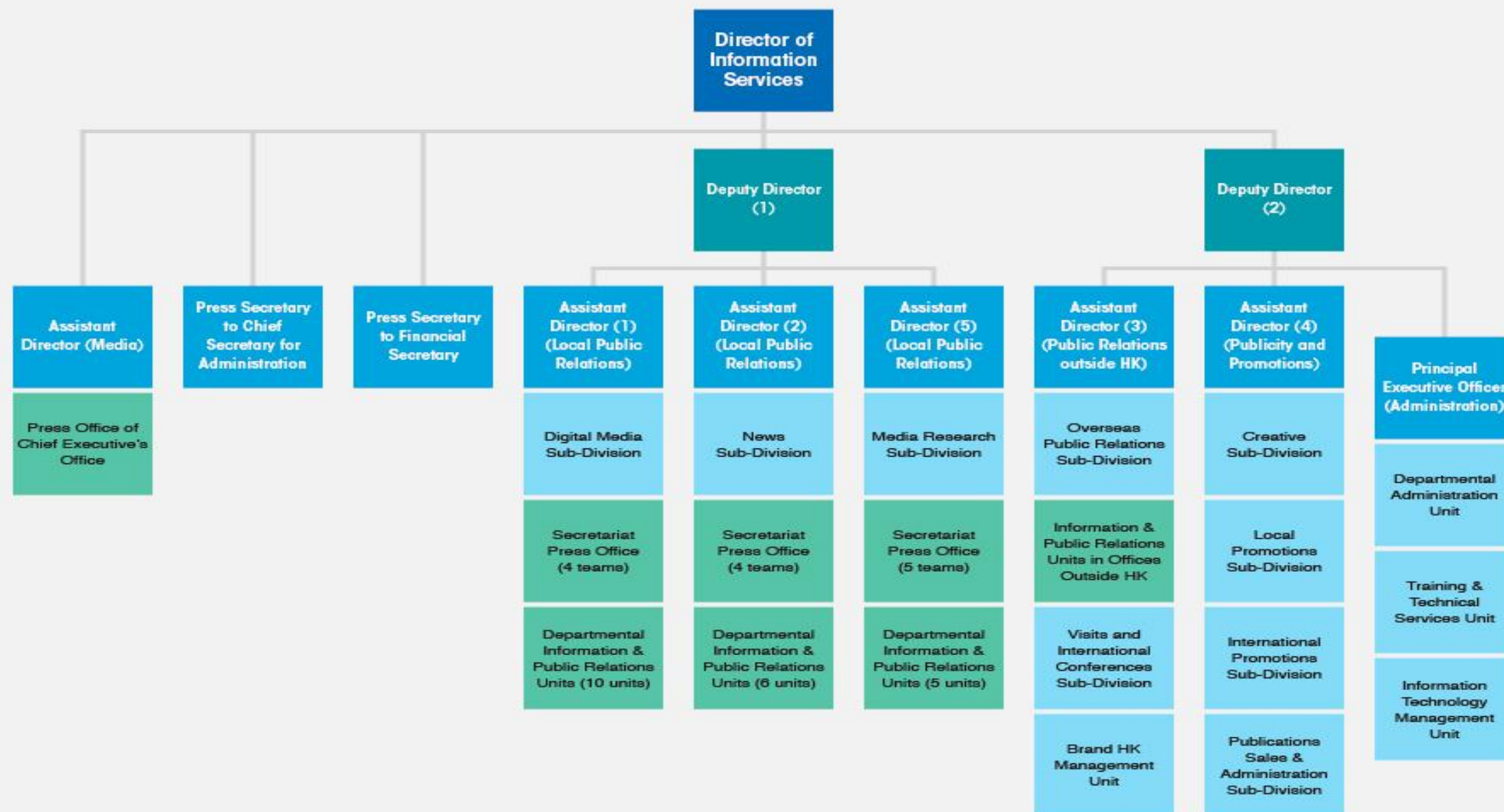


NASA organization and program chart.



Information Services Department - Organisation Chart

XX February 2018



Why one should study organization structure?

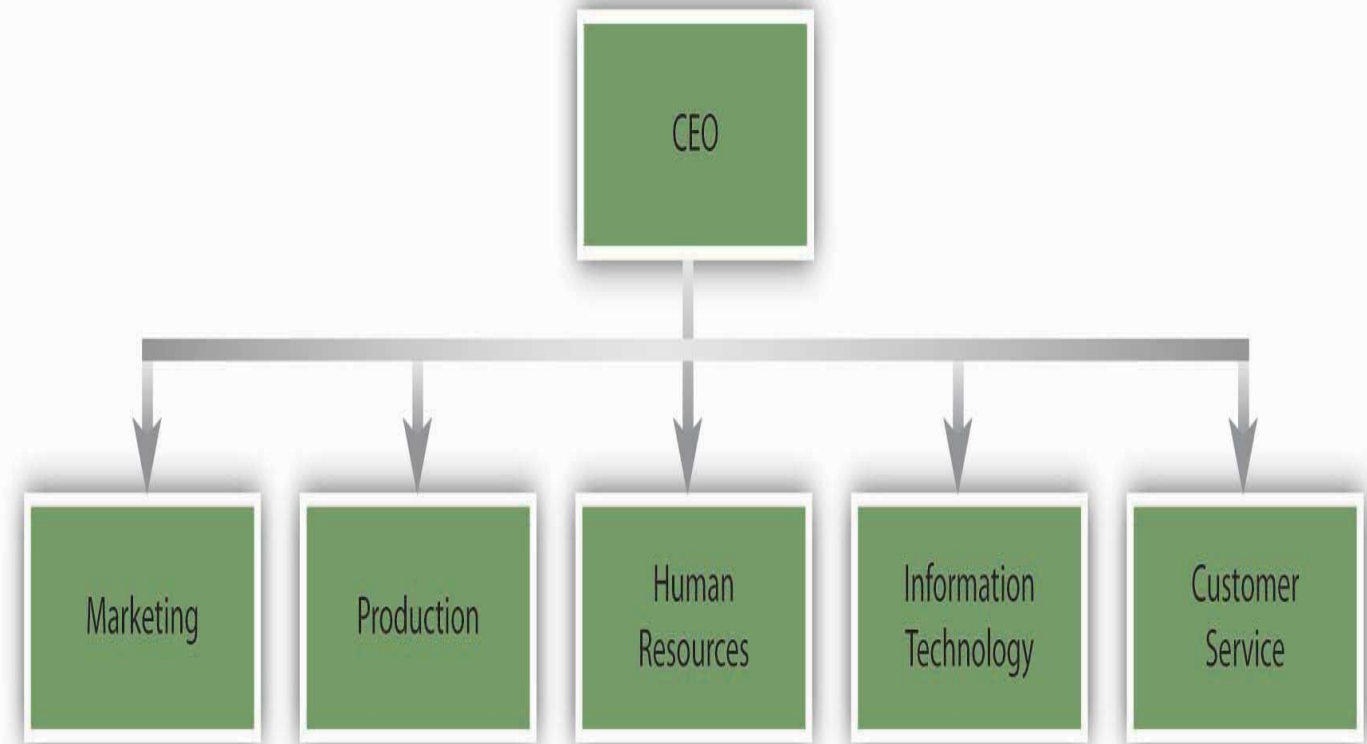
- ❖ Organization structure outlines how various roles, responsibilities, and tasks are assigned, coordinated, and controlled to achieve common goals and objectives
- ❖ The organizational structure is important, as it impacts every aspect of the company's operations, from its culture and strategic objectives to its day-to-day activities
- ❖ A well-designed and implemented organizational structure connects the employees to each other, and to their superiors on a clear chain of command, ensuring a smooth workflow process
- ❖ A well-designed and implemented organizational structure helps in creating an environment that fosters teamwork and collaboration, as well as promoting transparency and accountability

- ❖ A well-designed and effective organizational structure can also impact external stakeholders, such as clients and investors (who rely on the company's ability to deliver quality products or services in a timely and efficient manner)
- ❖ A well-organized company with a clear structure is better able to respond to-
 - changes in the market
 - take advantage of new opportunities
 - weather any challenges or crises that may arise
- ❖ A well-designed organizational structure can facilitate innovation and creativity, by providing a clear framework for the employees to work within, while also allowing them the autonomy to come up with new ideas and approaches

❑ A poorly structured organization can lead to confusion and chaos, ultimately resulting in decreased productivity, employee burnout, and a negative impact on the bottom line

Traditional/Classical/Functional organization:

- ❖ Traditional organizational structure is developed around the functional aspects of the organization like engineering, manufacturing, marketing, human resources, information systems
- ❖ The traditional management structure has survived for more than two centuries
- ❖ Recent business developments, such as the rapid rate of change in technology and increased stakeholder demands, have created strains on existing (Traditional) organizational forms
- ❖ For many companies survival depended on multiple product lines (i.e., diversification) and need of vigorous integration of technology into the existing organization
- ❖ Ineffective integration of activities in matured organization and new conflicts were arising in the well-established formal and informal channels



- ❑ Traditional organization structure works best in **small organizations** in which the different sections are geographically close together and which provide only a small number of goods and/or services

Advantages:

- Easier budgeting and cost control are possible
- Better technical control is possible
- Specialists can be grouped to share knowledge and responsibility
- Personnel can be used on many different projects.
- All projects will benefit from the most advanced technology (better utilization of scarce personnel)
- Flexibility in the use of manpower.
- A broad manpower base to work with.
- Continuity in the functional disciplines; policies, procedures, and lines of responsibility are easily defined and understandable
- Admits mass production activities within established specifications
- Good control over personnel, since each employee has one and only one person to report to
- Communication channels are vertical and well established
- Quick reaction capability exists, but may be dependent upon the priorities of the functional managers

Traditional/Classical/Functional organization:

Disadvantages:

- No one individual is directly responsible for the total project (i.e., no formal authority; committee solutions)
- Does not provide the project-oriented emphasis necessary to accomplish the project tasks
- Coordination becomes complex, and additional lead time is required for approval of decisions.
- Decisions normally favor the strongest functional groups.
- No customer focal point
- Response to customer needs is slow.
- Difficulty in pinpointing responsibility; this is the result of little or no direct project reporting, very little project-oriented planning, and no project authority
- Motivation and innovation are decreased.
- Ideas tend to be functionally oriented rather than project oriented

Pure Product (projectized) organization

- Projectized organization is a project focused organizational structure where project manager has the final authority over the project to make project decisions, priorities, acquire and assign resources.
- A **projectized organization** refers specifically to an organizational structure that has been set up in a manner in which the **project manager leads the group and in which the project manager has the ultimate authority** to make any and all decisions involving the organization.
- In a projectized structure, all the work is looked at as a project. The project manager has complete control, unlike in the functional structure, and all team members report directly to the project manager.
- The project manager works with **complete autonomy** (central authority, power) over the project, team is on the project **full time** and **report to the project** manager.

Project
Coordination

Chief
Executive

Projectized Organization

Project
Manager

Staff

Staff

Staff

Project
Manager

Staff

Staff

Staff

Project
Manager

Staff

Staff

Staff

(PMP 6th Edition)

(Gray boxes represent staff engaged in project activities.)

Pure Product (projectized) organization

Advantages:

- Provides complete line authority over the project (i.e., strong control through a single project authority).
- Participants work directly for the project manager. Unprofitable product lines are easily identified and can be eliminated.
- Strong communications channels.
- Staffs can maintain expertise on a given project without sharing key personnel.
- Very rapid reaction time is provided.
- Personnel demonstrate loyalty to the project; better morale with product identification.
- A focal point develops for out-of-company customer relations.
- Flexibility in determining time (schedule), cost, and performance trade-offs.
- Interface management becomes easier as unit size is decreased.
- Upper-level management maintains more free time for executive decision making.

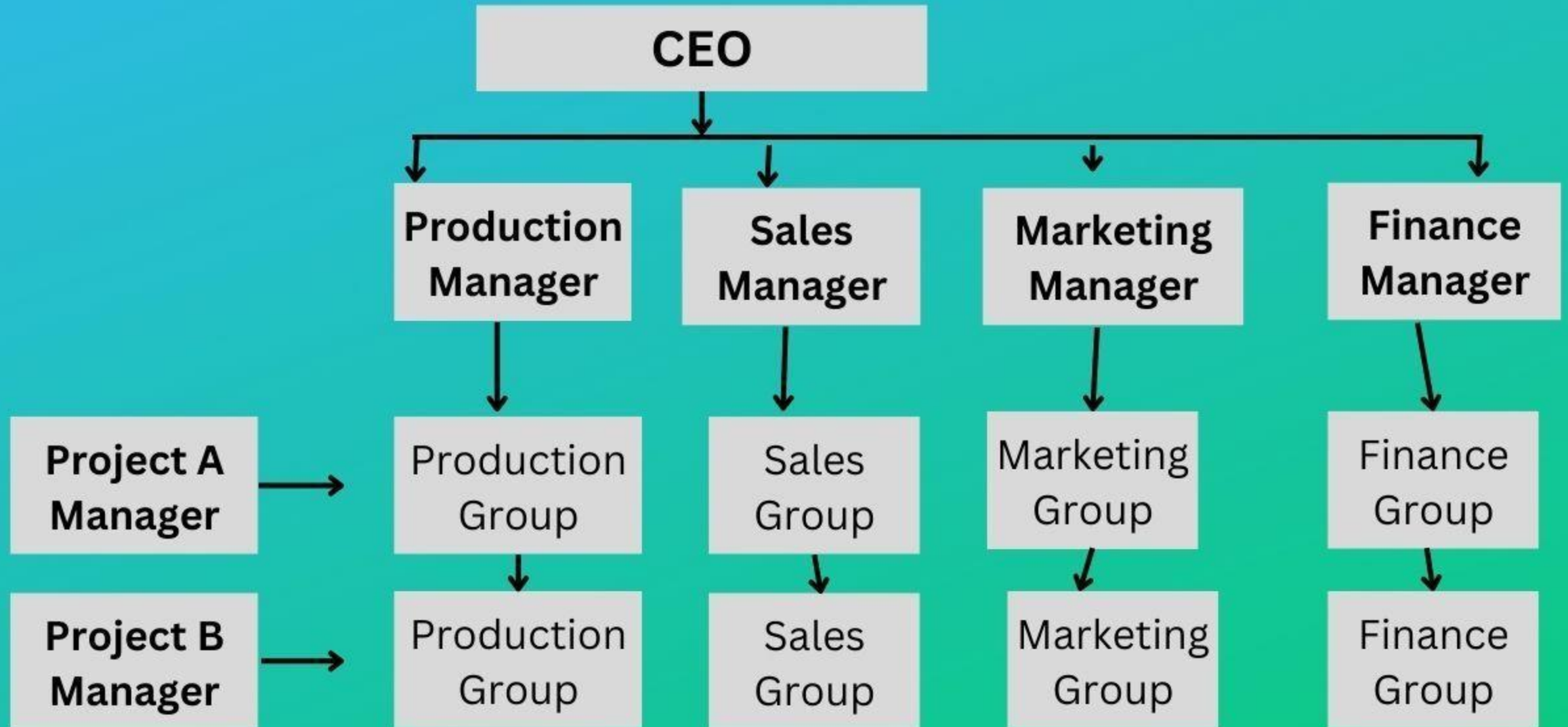
Disadvantages:

- Cost of maintaining this form in a multiproduct company would be prohibitive due to duplication of effort, facilities, and personnel; inefficient usage.
- A tendency to retain personnel on a project long after they are needed. Upper-level management must balance workloads as projects start up and are phased out.
- Technology suffers because, without strong functional groups, outlook of the future to improve company's capabilities for new programs would be hampered (i.e., no perpetuation of technology).
- Control of functional (i.e., organizational) specialists requires top-level coordination.
- Lack of opportunities for technical interchange between projects.
- Lack of career continuity and opportunities for project personnel.

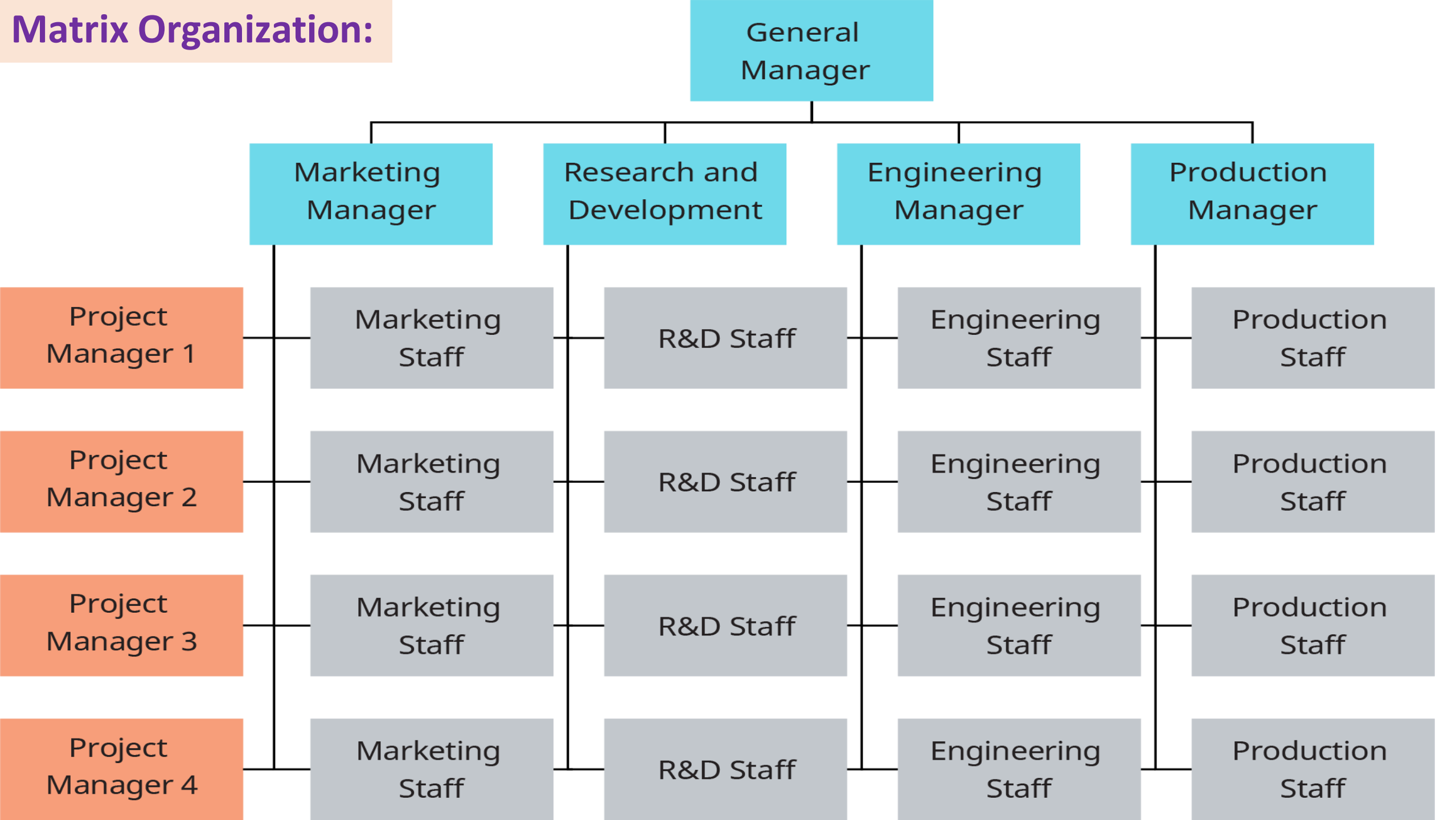
Matrix Organization:

- It is used mainly in the management of large projects or product development processes, drawing employees from different functional disciplines for assignment to a team without removing them from their respective positions.
- **Matrix management** is a type of organizational management in which people with similar skills are pooled for work assignments.
- The matrix structure **combines** both the functional and projectized structures. Each team member has two bosses; they report both to the functional manager and the project manager.
 - If the matrix is strong, the power resides more with the project manager.
 - If the matrix is weak, the power resides more with the functional manager.
 - The key is to find a balance in which the power is shared equally.
- Because of its **complexity**, this type of structure can lead to problems if it is not used carefully and properly. Good communication is essential for success.

Matrix Organizational Structure



Matrix Organization:



Matrix Organization:

Advantages:

- The project manager maintains maximum project control (through the line managers) over all resources, including cost and personnel.
- Policies and procedures can be set up independently for each project, provided that they do not contradict company policies and procedures.
- The project manager has the authority to commit company resources, provided that scheduling does not cause conflicts with other projects.
- Rapid responses are possible to changes, conflict resolution, and project needs (as technology or schedule).
- The functional organizations exist primarily as support for the project.
- Each person has a “home” after project completion. People are susceptible to motivation and end-item identification. Each person can be shown a career path.

- Because key people can be shared, the program cost is minimized. People can work on a variety of problems; that is, better people control is possible.
- A strong technical base can be developed, and much more time can be devoted to complex problem solving. Knowledge is available for all projects on an equal basis.
- Conflicts are minimal, and those requiring hierarchical referrals are more easily resolved.
- There is a better balance among time, cost, and performance.
- Rapid development of specialists and generalists occurs.
- Authority and responsibility are shared.
- Stress is distributed among the team (and the functional managers).



Matrix Organization:

Disadvantages:

- Multidimensional information flow.
- Multidimensional work flow.
- Dual reporting.
- Continuously changing priorities.
- Management goals different from project goals.
- Potential for continuous conflict and conflict resolution.
- Difficulty in monitoring and control.
- Company-wide, the organizational structure is not cost-effective because more people than necessary are required, primarily administrative.
- Each project organization operates independently. Care must be taken that duplication of efforts does not occur
- More effort and time are needed initially to define policies and procedures, compared to traditional form.

- Functional managers may be biased according to their own set of priorities.
 - Balance of power between functional and project organizations must be watched.
 - Balance of time, cost, and performance must be monitored.
 - Although rapid response time is possible for individual problem resolution, the reaction time can become quite slow.
 - Employees and managers are more susceptible to role ambiguity than in traditional form.
 - Conflicts and their resolution may be a continuous process (possibly requiring support of an organizational development specialist).
 - People do not feel that they have any control over their own destiny when continuously reporting to multiple managers.
- 

Ground rules for matrix form of development:

- Participants are full time on the project; this ensures a degree of loyalty.
- Horizontal as well as vertical channels must exist for making commitments.
- There must be quick and effective methods for conflict resolution.
- There must be good communication channels and free access between managers.
- All managers must have input into the planning process.
- Both horizontally and vertically oriented managers must be willing to negotiate for resources.
- The horizontal line must be permitted to operate as a separate entity except for administrative purposes.

Requirements of Project Organizations:

- ❖ Project environments are characterized by
 - Complexity
 - Change,
 - Uncertainty
- ❖ Projects typically require the resources and coordinated work effort of multiple people, subunits, and organizations
- ❖ Changes, mistakes, or delays in one subunit have consequences for all others
- ❖ Organizations working in technologies such as software development, pharmaceuticals, biomedicine, space exploration, product development encounter the unexpected
- ❖ Above type of companies need to be adaptable to
 - changing goals
 - customer needs
 - environmental demand
 - resources,
- ❖ Be able to deal with the uncertainty that accompanies the above changes

Selecting an Organization form/structure for Project:

- ❖ It is difficult to state which organization form is always best
- ❖ Criteria to help/guide for selection-
 - Frequency of new projects (how often, or to what degree the parent company is involved in project-related activity)
 - Duration of projects (how long a typical project lasts)
 - Size of projects (level of human, capital, or other resources in relation to other activities of the company)
 - Complexity of relationships (number of functional areas involved in the project and degree of interdependency, including transparency at the senior management level)
 - The experience of the organization in managing projects
 - The uniqueness of the project

- Changing from a traditional structure to a project management structure is a big leap forward
- Once project management structure is adopted, it is not possible to revert back to traditional structure

Issues in Project Organization*:

The firm has to answer the following questions while making the choice of project organization:

- How much independence should be given to the project team?
- Where should a project organization fit in the total organization?
- How should people be assigned to the project?
- What directive/authority should managers have (i.e., degree of empowerment)?



**GREAT THINGS IN BUSINESS ARE
NEVER DONE BY ONE PERSON.
THEY'RE DONE BY A TEAM OF
PEOPLE.**

- Steve Jobs

Project management includes:

- **Project Manager**

- Assistant project managers if necessary
- A project (home) office/Project Support office
- A project team

Project Team is defined* as a group of people (small number of people) with complimentary skills who are committed to a common purpose, performance, goals and approach who are directly or indirectly accountable to project manager

* Reference: “*Project Management for IT-Related Projects*”, Bob Hughes, Roger Ireland, Brian West, Norman Smith, David Shepherd, The British Computer Society, 2007

Project Manager:

- ❖ The project manager is the centerpiece of project management
- ❖ A project manager is a professional who organizes, plans, and executes projects while working within restraints like budgets and schedules
- ❖ They are responsible for managing people, tasks, schedules, and resources through the project life cycle
- ❖ Project managers lead entire teams, define project goals, communicate with stakeholders, and see a project through to its closure
- ❖ *Project manager's prime role is to integrate everything and everybody to accomplish project goals*

“Project managers are organized, goal-oriented professionals who use innovation, creativity, and collaboration to lead projects that make an impact”- PMI

❑ Job Responsibilities*-

- Planning project tasks and end results, creating the WBS (work breakdown structure), schedule, budget, coordinating tasks, allocating resources
- Selecting & Organizing the project team
- Working and negotiating with influential stakeholders (customers, functional managers, contractors, supporters and top management) and managing their expectations
- Monitoring project status and communicating status to stakeholders
- Identifying technical and functional problems
- Solving problems directly or knowing where to find help
- Dealing with crises and resolving conflicts

*J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology-Principle & Practice*, 5th Edition , Routledge (Taylor & Francis), 2019

Project Manager:

- Projects usually involve complex and multifaceted tasks, requiring the skills of a variety of people
- Variety of skills project manager should possess to coordinate the work of all the team members and successfully complete the project

Skills set required for Project Manager* (qualifications of project manager**):

1. *Team Building*
2. *Leadership*
3. *Conflict resolution*
4. *Technical expertise*
5. *Planning*
6. *Organization*
7. *Entrepreneurship*
8. *Administration*
9. *Management support*
10. *Resource allocation*

1. *Team - Building Skill*

- ❑ Team building is the main responsibility of the project manager
- ❑ This involves integrating different work groups from functional groups into a project management system
- ❑ The project manager should provide a comfortable work environment and should nurture following characteristics
 - Team members committed to the project
 - Good interpersonal relations and team spirit
 - The necessary expertise and resources
 - Clearly defined goals and project objectives
 - Involved and supportive top management
 - Good project leadership
 - Open communication among team members and support organizations
 - A low degree of detrimental interpersonal and intergroup conflict
- ❑ To have good/comfortable work environment major considerations
 - i. Effective communications
 - ii. Sincere interest in the professional growth of team members
 - iii. Commitment to the project

• References: * H. Kerzner, Project Management, 12th edition, Wiley ; * PMBOK ® Guide, 6th Edition

**J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology-Principle & Practice, 5th Edition* , Routledge (Taylor & Francis), 2019

2. Leadership

- ❑ A prerequisite for project success is the project manager's ability to lead the team within a relatively unstructured environment
- ❑ Leadership is defined as a process of social influence, which maximizes the efforts of others, towards the achievement of a goal
- ❑ Leadership involves dealing effectively with managers and supporting personnel across functional lines
- ❑ Leadership is ability to collect and filter relevant data for decision making in a dynamic environment
- ❑ Leadership is ability to integrate individual's into decisions
- ❑ Leadership is ability to resolve intergroup conflicts

Demands

Requirements

Limitations

Table*: Three aspects of Leadership and Fifteen Leadership Competencies

Area of Competence	Competency
<i>Intellectual (IQ)</i>	1. Critical analysis and judgment 2. Vision and imagination 3. Strategic perspective
<i>Managerial (MQ)</i>	4. Engaging communication 5. Managing resources 6. Empowering 7. Developing 8. Achieving
<i>Emotional (EQ)</i>	9. Self-awareness 10. Emotional resilience 11. Motivation 12. Sensitivity 13. Influence 14. Intuitiveness 15. Conscientiousness

Reference: *1. Text book one from syllabus

*2. Dulewicz, V., and M. J. Higgs. "Design of a new instrument to assess leadership dimensions & styles." In Henley Working Paper HWP 0311, Henley Management College, Henley-on-Thames, UK. 2003

**3. Müller, Turner. "Leadership Competency Profiles of Successful Project Managers." International Journal of Project Management, Vol 28, 2010.

- **Engineering including construction projects, IT projects, Organizational change projects all required different levels of 15 competencies to be successful
- Competency levels required varied with; 1. Complexity of the project, 2. Importance of the project, 3. Type of contract under which project is carried out

3. Conflict resolution

- ❑ Conflict is fundamental to complex task management
- ❑ Conflict resolution is essential for the effective management of complex tasks
- ❑ By minimizing the conflicts, the project manager enhances the performance of the project team
- ❑ To resolve the conflicts the project manager must-
 - Analyze the organizational and behavioral interaction of teams to create an environment that matches the motivational needs of the teams, the idea is to minimize conflicts & improve cooperation
 - Ensure that the project objectives & decisions are communicated across all organizational levels
 - Identify/Recognize the conflict determinants along with their frequency (timings) occurrence in the project life cycle
- ❑ Conflicts can be minimized by involving senior management & contingency planning
- ❑ Project manager should encourage conflicts that generate new ideas/solutions & discourage conflicts that can be detrimental to the project

- When conflict becomes dysfunctional, it often results in-
 - Poor project decision making
 - Lengthy delays over issues
 - Disruption of the team's efforts
 - all negative influences to project performance
- Conflicts can be beneficial-
 - When it produces involvement
 - New information
 - Enhances the competitive spirit

4. Technical expertise/Technical Skills

- ❑ The project manager rarely has all the technical, administrative, and marketing expertise needed to direct the project single-handedly
- ❑ To initiate new projects, the project manager must understand technology, analyze the markets and business environment
- ❑ Lack of technical skills adversely affects the performance of technical manager
- ❑ Technical expertise is necessary for project manager-
 - To evaluate technical concepts and solutions
 - To communicate effectively in technical terms with the project team, senior management
 - To assess risks and make trade-offs between cost, schedule, and technical issues
- ❑ Project manager with technical expertise should be able to understand-
 - The technology involved in the project
 - The engineering tools, techniques and procedures used
 - The requirements of the customers in specific markets
 - The applications of the product
 - Evolving technological trends
 - Supporting technologies and their relationships
- ❑ In non or low technology projects, grasp can be developed through experience and informal training
- ❑ In high-technology projects, technical qualifications are more rigorous, requires career (of project manager) molded in the technology environment
- ❑ For high-technology projects a degree in science/engineering is important

5. Planning Skills

- ❑ Planning skills are absolutely essential for the successful management of large complex projects
- ❑ The project plan is the road map that defines how to get from the start to the final results
- ❑ Preparation of a project summary plan, prior to project start, is the responsibility of the project manager
- ❑ Effective planning requires the ability to-
 - Process information
 - Communicate
 - Negotiate for resources and administrative support
 - Secure commitments from management
 - Plan in an incremental and modular fashion
 - Ensure participation of top management
- ❑ Adjustments for changes in project scope and depth
- ❑ Plan should be the guiding document throughout the lifecycle of the project

6. Organizing Skills

- ❑ These skills very important & significant when designing & initiating the project
- ❑ The project manager must be a social architect
- ❑ Project Manager must understand how the organization works and how to work with the organization
- ❑ This skill involves describing-
 - Reporting structure
 - Responsibilities
 - Line of control
 - Information needs
- ❑ This skill becomes an effective tool when supplemented with-
 - Clearly defined project objectives/goals
 - Proper & open communication channels
 - Good project leadership
 - Senior management support

7. Entrepreneurial Skills

- ❑ The project manager needs a general management perspective also
- ❑ Should have concern for –
 - Customer relationship & satisfaction
 - Cultivation of related market activities
 - Minimum organizational disruptions
 - Future growth prospects
- ❑ These skills can be developed more profoundly through real time experience
- ❑ Formal MBA-type training, special seminars, and cross-functional training projects can help the project managers to develop the entrepreneurial skills

8. Administration/Administrative Skills

- ❑ Project manager must have administrative skills
- ❑ He must be able to do planning, staffing, budgeting, scheduling and use of controlling techniques
- ❑ Project managers must be thoroughly familiar & know how to use effectively following helpful administrative tools-
 - (1) the meeting, (2) the report, (3) the review, and
 - (4) the budget and schedule controls

9. Management support/ Supporting Management Skills

- ❑ These skills enhance the chances of the project manager in building relationship with top management
- ❑ Only a strong leader (project manager) backed by senior management can prevent the development of unfavourable biases and resolve the conflicts

10. Resource allocation skills

- ❑ It involves designing a **work statement** (which establishes a basis for resource allocation)
- ❑ The project manager is also responsible for organizing meetings with key vendors (including their senior officials) to discuss tasks, budgets, delivery schedules (of vendor's goods)

EI : Emotional Intelligence -Project Manager

- EI is defined *as the ability to monitor one's own and others' feelings and emotions to distinguish between them, and to use this information to guide thought and action
- EI involves the evaluation & expression of emotions, emotional regulation, and the use of emotional content in conflict resolution
- EI involves the regulation and management of one's own emotions, as well as of others, to achieve desired outcomes
- Daniel Goleman's model** breaks EI into five key components:

Self-awareness

Self-regulation

Motivation

Empathy

Social skills

- PMI states that project managers need to balance technical & interpersonal skills to analyze and interact effectively in different situations
- Research shows that EI makes a positive contribution to the competencies that are considered to be fundamental in Project management, especially communication & conflict resolution
- Investment in developing the emotional & social skills of project managers in any area that involves interdisciplinary work and/or contact with stakeholders is important, as these directly influences the success of project and team cohesion
- Project Managers' higher EI results in better relationship with internal and external stakeholders helping to make project successful

Note: *Today, Effective project management is not just technical skills; it also incorporates an emotional dimension that plays a decisive role in the success and dynamics of teams*

Experience –Project Manager:

- The **experience of a project manager** is often valued more than educational qualifications in **large or mega projects** due to the **complexity, scale, and unpredictability** involved

1. Real-World Problem Solving

- Mega projects face unexpected challenges—technical issues, stakeholder conflicts, regulatory hurdles, etc.
- Experienced PMs have practical exposure to such situations and can respond with proven strategies

2. Leadership Under Pressure

- Large projects involve high stakes, tight deadlines, and intense scrutiny
- Experience builds resilience, decision-making maturity, and the ability to lead under pressure—qualities not easily taught in classrooms

3. Stakeholder Management

- Mega projects often involve multiple stakeholders: government bodies, clients, vendors, communities.
- Experienced PMs know how to navigate politics, build consensus, and manage expectations effectively

4. Risk and Change Management

- Big projects are more prone to scope creep, budget overruns, and regulatory changes
- A seasoned PM can anticipate risks, develop contingency plans, and adapt quickly

5. Team and Resource Coordination

- Managing large teams across geographies and disciplines requires strong interpersonal and organizational skills
- Experience helps in delegating tasks, resolving conflicts, and maintaining productivity

6. Track Record of Delivery

- Top management prefers PMs with a history of successful project completions.
- Past performance is a reliable indicator of future capability, especially in high-investment projects.

Experience –Project Manager:

Why Educational Qualification Alone Isn't Enough???

- Education provides theoretical knowledge, but not the practical wisdom needed for complex project environments.
- It may not prepare PMs for real-time decision-making, human dynamics, or political challenges.
- In mega projects, experience is not just an asset—it's a necessity ,While educational qualifications are important for foundational knowledge, hands-on experience equips project managers with the tools to lead, adapt, and deliver in high-stakes environments

- ❖ **Note*:** Larger the project size, the greater the like hood that the organization will seek an older, more experienced project manager
- ❖ Experience is made up of two component, one project experience and second total experience

Four Types of Project Mangers:



SOURCE CARSTEN PEDERSEN AND THOMAS RITTER © HBR.ORG

Which Type of Project Manager Is Right for Which Project?

	PROPHET	GAMBLER	EXPERT	EXECUTOR
Strategy should be...	Challenged	Followed	Challenged	Followed
The growth opportunity is...	A grand vision	A bet	An analysis	A sure thing
Organizational followers should...	Make a leap of faith	Gamble a bit and pursue rewards	Listen to the advice and act upon it	Stay within the strategy and follow the analyses
Weapon of choice	Persuasive vision	Potential reward	Well-supported arguments	Business cases and reports

SOURCE CARSTEN PEDERSEN AND THOMAS RITTER © HBR.ORG

Four Types of Project Mangers:

Prophet-

- ❑ This type of project manager actively pursues business opportunities that lie outside the existing strategic boundaries in an area where it is extremely difficult to obtain trustworthy data concerning the likelihood of success
- ❑ This type of project manager seeks to gain organizational followers for a grand vision of a growth opportunity that is strategically different from the status quo — and without trustworthy quantitative evidence, consequently relying on organizational members making a leap of faith in support of the vision

Gambler-

- ❖ This type of project manager actively pursues business opportunities that lie within the existing strategic boundaries but have no good business case attached, as trustworthy data concerning the likelihood of success is lacking
- ❖ This type of project manager seeks to gain organizational followers for a big bet on a growth opportunity that is consistent with the current strategy but without trustworthy quantitative evidence

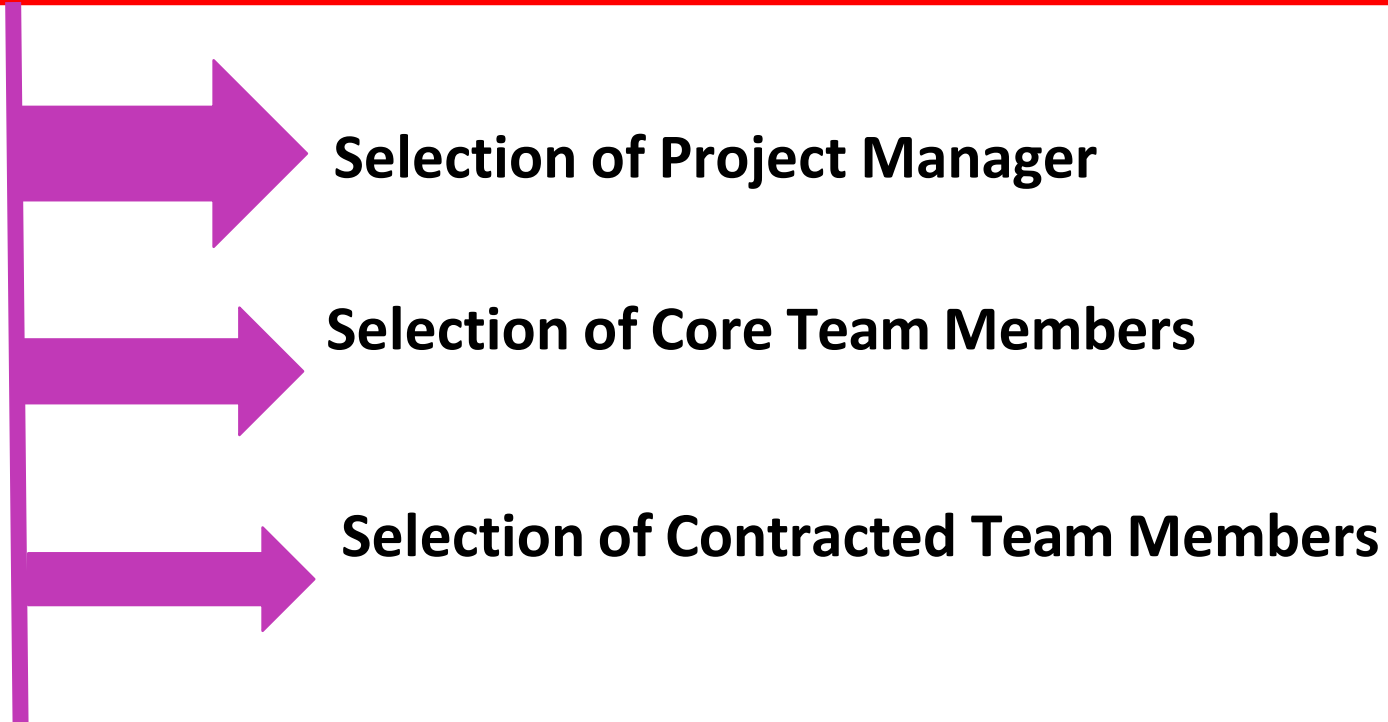
Expert-

- This type of project manager actively pursues business opportunities that lie outside the existing strategic boundaries but for which trustworthy data builds a solid business case
- This type of project manager wish to gain organizational followers for a change in action in favour of a growth opportunity that is inconsistent with the current strategy but is supported by solid, trustworthy quantitative evidence

Executor-

- This type of project manager actively pursues business opportunities that lie within the existing strategic boundaries and have great cases
- This type of project manager gains organizational followers for a sure-thing growth opportunity that is consistent with the current strategy and is backed by trustworthy quantitative evidence

THE ORGANIZATIONAL STAFFING PROCESS/ SELECTING THE STAFF REQUIRED



Selection of Project Manager:

- ❖ Probably the most difficult decision facing upper-level management is the selection of project managers
- ❖ Generally, the project organization establishes a committee to screen candidates for the project manager
- ❖ This committee-
 - Establishes a selection criteria along with the professional qualifications of the project manager
 - Acts according to the selection policy received from top management
 - Involves senior management in the selection process
- ❖ List of the parameters for screening candidate-
 - Background and experience
 - Leadership and strategic expertise
 - Technical expertise
 - People management skills

Reference: Project Management – ICFAI Center for Management Research

- ❑ The selection of project managers is based more on personal characteristics than on the job description
- ❑ Broader range of desired personal characteristics for Project Manager * -
 - Flexibility and adaptability
 - Preference for significant initiative and leadership
 - Aggressiveness, confidence, persuasiveness, verbal fluency
 - Ambition, activity, forcefulness
 - Effectiveness as a communicator and integrator
 - Broad scope of personal interests
 - Poise, enthusiasm, imagination, spontaneity
 - Able to balance technical solutions with time, cost, and human factors
 - Well organized and disciplined
 - A generalist rather than a specialist
 - Able and willing to devote most of his time to planning and controlling
 - Able to identify problems
 - Willing to make decisions
 - Able to maintain proper balance in the use of time

*Russell D. Archibald, Managing High-Technology Programs and Projects (New York: John Wiley & Sons,1976)

Selecting Core Team Members:

- ❖ Recruiting and selecting the core team members (functional team members) is a challenging task for the project manager
- ❖ Selecting core team members at the beginning of the project initiation phase helps their participation in the planning of the project
- ❖ The expectations from core team members-
 - Commitment to the project
 - Sharing responsibility
 - Adaptability
 - Orientation towards the task
 - Staying on schedule
 - Trustworthiness and supportiveness
 - Orientation towards team
 - Open-mindedness
 - Flexibility to function across the structure and authority
 - Usage of project management tools

Selecting Contracted Team Members:

- ❖ Organizations outsource as they do not have sufficient staff or lack the expertise (means they have to hire external people to work on project for limited period of time)
- ❖ E.g. – Many software development firms source their project team members on contract basis
- ❖ Project manager need to consider following points for recruitment of good contract team/person
 - Determine the skill set needed and the number of personnel required
 - Identify the list of companies/contractors/manpower consultancies to be invited to submit proposals
 - Set up the criteria to analyze the proposals and select company/contractors/manpower consultant
 - Enter into the contract after choosing company/contractors/manpower consultant

❑ *Problems arises due to contracted team members-*

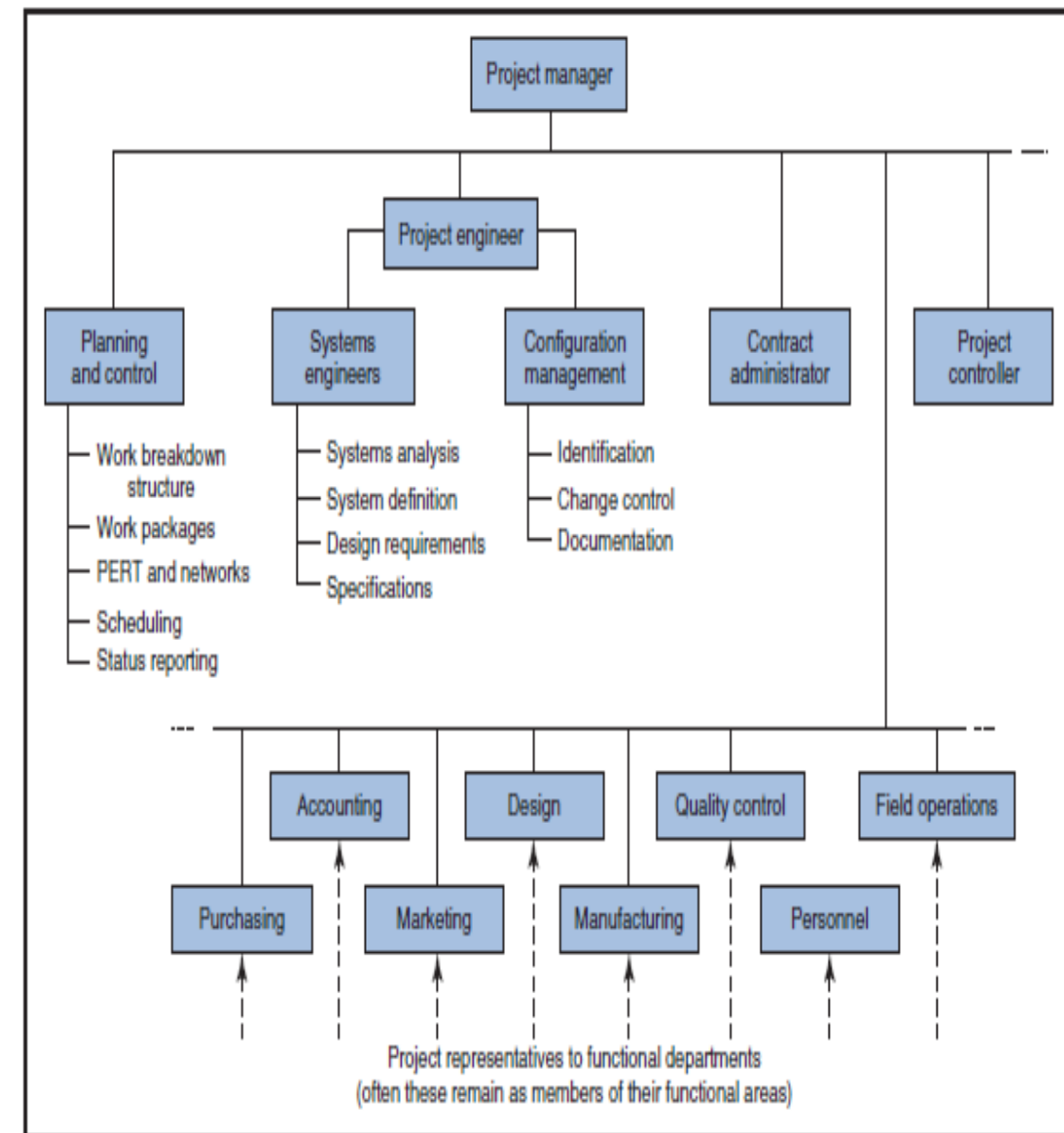
- As they work for short period of time, their activities must be carefully integrated with the project schedule
- They have to be assigned a role in the project and their relationship with other activities must also be defined
- They may have a low level of commitment and quality standards may be violated
- They may require greater supervision and monitoring and the project manager will be forced to take on this additional responsibility

Reference: 1. Project Management – ICFAI Center for Management Research

2. D. N. Antoniadis, *Complexity and the Process of Selecting Project Team Members*, Journal for the Advancement of Performance Information and Value VOL. 4 NO. 1, 2012

Project Office:

- ❖ The project team is a combination of the project office and functional employees
- ❖ The term *project office* has dual meaning: it refers first to a support staff group that reports to the project manager, and second to a physical place where the project team meets
- ❖ The functions and composition of the project office depend on the authority of the project manager, and the size, importance, and goal of the project
- ❖ The expected functions/responsibilities of Project Office-
 - To act as a center of information to both external and internal stakeholders
 - To follow contract requirements by controlling time ,cost and performance
 - To document the tasks performed and communicate the same to all stakeholders
 - To ensure authorization of tasks accomplished and their funding through contract documents
 - To integrate work across functional departments of the organization
- ❑ The project manager should employ only those individuals in project office who are interested in making career in project management
- ❖ When office coordinate multiple projects, staff is larger then called as Project Management Office (PMO)



Project office for a large development project

Reference: 1. Project Management – ICFAI Center for Management Research
2. J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology-Principle & Practice*, 5th Edition , Routledge (Taylor & Francis), 2019

Project Team Building:

- ❑ **Project Team** is defined as a group of people (small number of people) with complimentary skills who are committed to a common purpose, performance, goals and approach who are directly or indirectly accountable to project manager
 - ❑ Building your best team takes time and consideration
 - ❑ The need for a team building depends on the team and the nature of task
 - ❑ Every project is unique and requires expertise in many areas to pull it all together into a successful end goal and product that will make your company and any client proud
 - ❑ Note, more varied the backgrounds and responsibilities of team members the grater the need of team building
 - ❑ Projects involving innovation, new and advanced technologies, high risks, and tight schedules place teams under heavy stress (some stress will motivate the team, but too much is detrimental)
 - ❑ Team building can help team to deal with the stress and to disclose and resolve problems as they occur
 - ❑ The norms that team can strive in fulfilling the purpose of team building i.e. enhance group's ability to work together-
1. *Effective communication among members.*
 2. *Effective resolution of group process problems.*
 3. *Techniques for constructively using conflict.*
 4. *Greater collaboration and creativity among team members.*
 5. *A more trusting, supportive atmosphere within the group.*
 6. *Clarification of the team's purpose and the role of each member*
- ❑ Make sure every seat at the project table is occupied with the best possible choice for the challenge, and keep them engaged and encouraged to perform their best

Project Team Building:

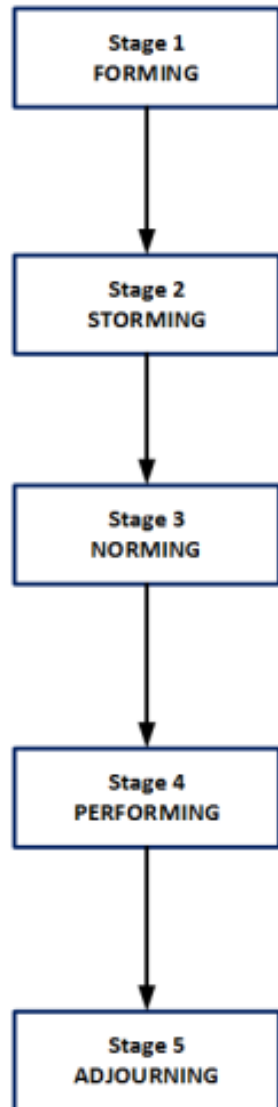


Exhibit 1: 5 Stages

- Tuckman Model* for team building describes five phases/stages through which every team passes including project team

Stage 1: Forming

Stage 2: Storming

Stage 3: Norming

Stage 4: Performing

Stage 5: Adjourning

- This model helps project manager in following ways-

1. Helps the project manager anticipate team behavior

You know when to expect conflicts, confusion, or high performance.

2. Provides guidance on leadership style

Each stage demands different leadership (directive → coaching → supportive → delegating).

3. Supports risk and stakeholder management

Understanding group dynamics reduces:

- communication breakdown
- role misunderstandings
- morale issues

4. Enhances team performance and project success

When used well, the model helps accelerate movement toward the Performing stage

Reference: 1. * The Five Stages of Team Development: A Case Study By Gina Abudi (Published by PMI as white paper)

(file:///C:/Users/Asus/Downloads/The_Five_Stages_of_Team_Growth_A_Mini_Case_Study.pdf)

2. B W Tuckman (1965), 'Developmental Sequence in Small Groups', Psychological Bulletin 63.

3. B W Tuckman and M A C Jensen (1977), 'Stages of small group development revisited', Group and Organization Studies, vol.2, no.4, pp.419-27.

4. "Project Management for IT-Related Projects", Bob Hughes, Roger Ireland, Brian West, Norman Smith, David Shepherd, The British Computer Society, 2007, pp.118-119

Integration in Large Scale Projects:

Mega projects, sometimes called “major programs”, are large-scale, complex ventures that typically cost more than 1 billion US Dollars, take many years to build, involve multiple public and private stakeholders, are transformational, and impact millions of people

- ❑ “Project Integration Management includes the processes and activities to identify, define, combine, unify, and coordinate the various processes and project management activities within the Project Management Process Groups.”- A Guide to the Project Management Body of Knowledge (PMBOK 6)
- ❑ The most difficult aspect of implementing the plan for a complex project is the coordination and integration of the various elements of the project so that they meet their joint goals of scope, schedule, and budget in such a way that the total project meets its goals
- ❖ Most Large Scale Projects are devoted to development and/or construction of complex systems
- ❖ In a large-scale project (LSP) or *mega project*, numerous parties—sponsors, prime contractors, subcontractors, consultants, and suppliers—contribute to one effort/one goal

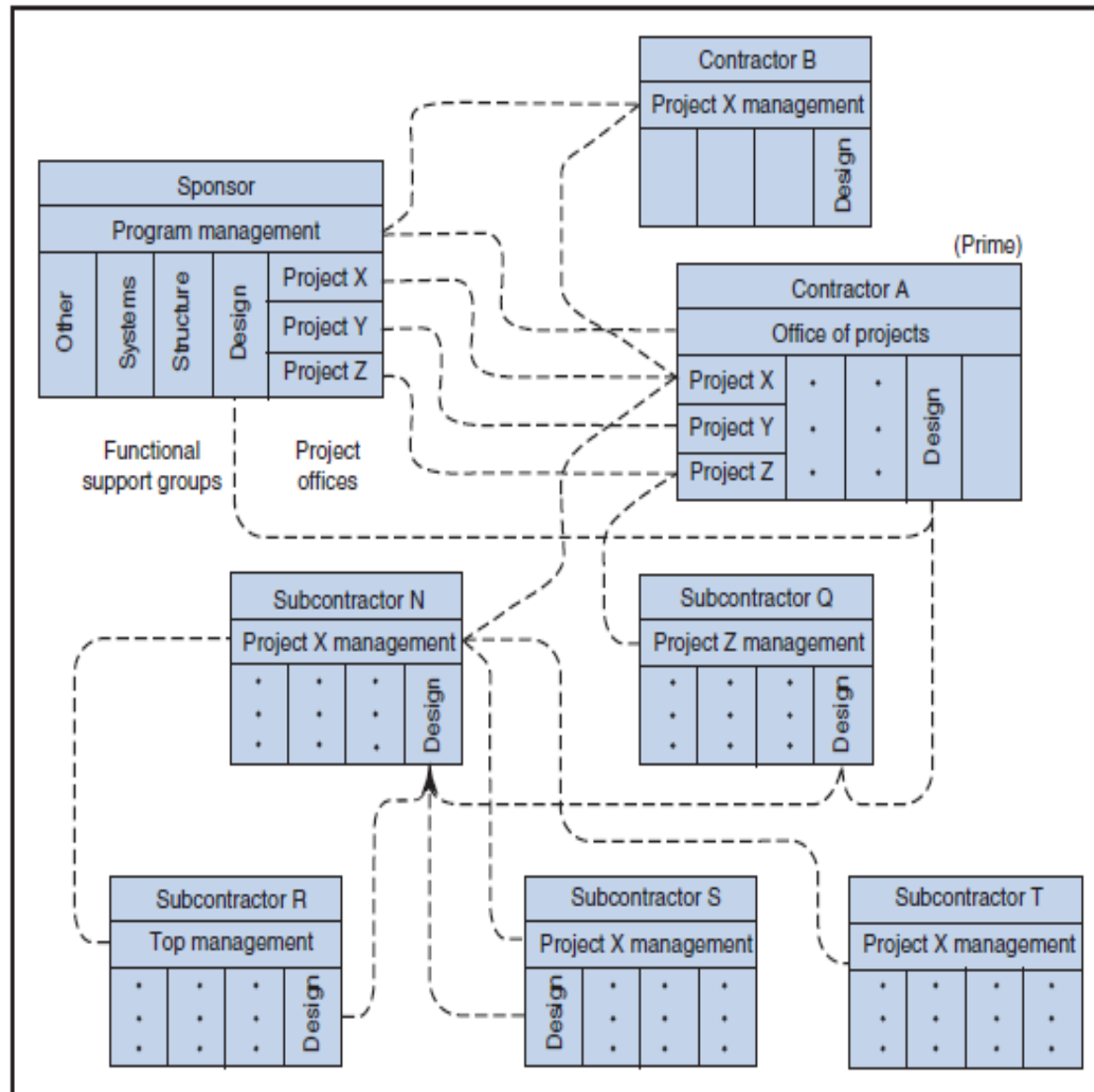
Affects these four dimensions

- ❑ Four dimensions* of successes for Mega or Large Scale Projects-
- 1. Efficiency (to what extent did the megaproject fulfill its cost, time, and other productivity goals?)
- 2. Impact on customers/users
- 3. Business/ Financial successes
- 4. Impact on Society

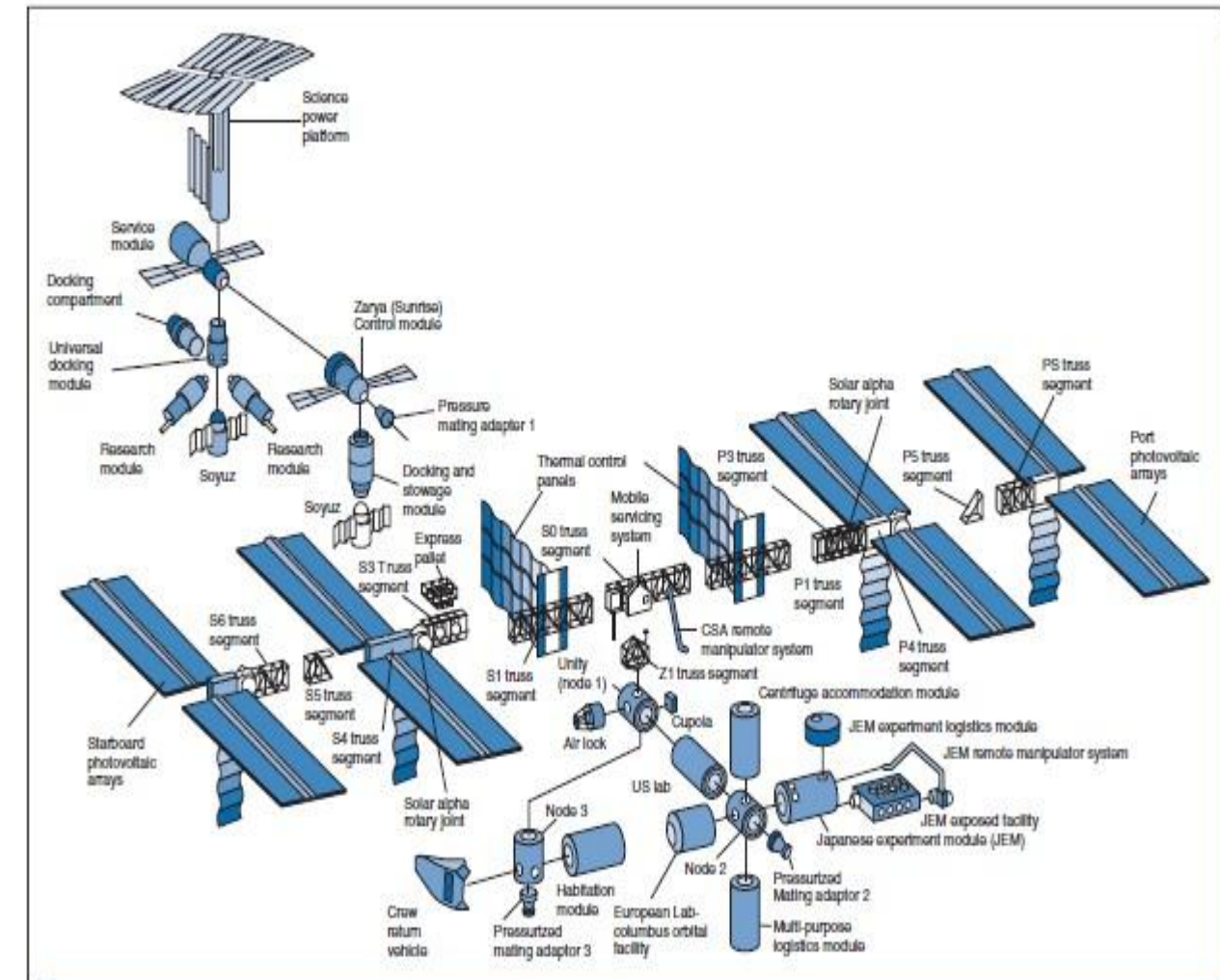
Reference: *Aaron Shenhar, Vered Holzmann, “*The Three Secrets of Megaproject Success: Clear Strategic Vision, Total Alignment, and Adapting to Complexity*”, Project Management Journal, Vol. 48, No. 6, 29–46 © 2017 by the Project Management Institute

Note: For more information on Integration in Large Scale Projects, Read- J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology- Principle & Practice*, 5th Edition, Routledge (Taylor & Francis), 2019

Integration relationships in a large-scale project.



Major components in the hardware and assembly of the International Space Station. Diagram courtesy of NASA.



For additional information read: J. Nicholas, H. Steyn, *Project Management for Business, Engineering & Technology-Principle & Practice, 5th Edition*, Routledge (Taylor & Francis), 2019

Questions from Previous Exam Question Papers

–MPSTME, Unit-Two

Unit-2	Project Organization Project organization, types of structures, roles of project managers and team, integration in large scale projects.	05 Hours
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CO-1; SO-4; BL-2 b. Compare and contrast the functional, matrix, and projectized project organization structures. [5]

CO-1; SO-7; BL-1 b. Define the roles and responsibilities of project managers and project teams. How do they contribute to the success of a project? [10]

CO-1; SO-4; BL-2 b. Explain the roles and responsibilities of a project manager in a project organization. [5]

CO-1; SO-7; BL-2 b. Differentiate between various types of project organization structures. Provide examples of when each structure is most suitable.- [10]

CO-1; SO-4; BL-2	a.	Explain the role of project manager and its team in detail.	[05]
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CO-1; SO-4; BL-2	a.	Explain the role of project manager in detail and also discuss interpersonal skills that project manager must possess.	[10]
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CO-1; SO-4; BL-2	a.	Discuss the interpersonal skills required to be a successful project manager and describe how each skill contributes to a project manager's success.	[10]
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CO-1; SO-1; BL-2	a.	Describe the role of structure of the organization in project management.	[10]
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A Explain types of project Managers and their roles.

(7)

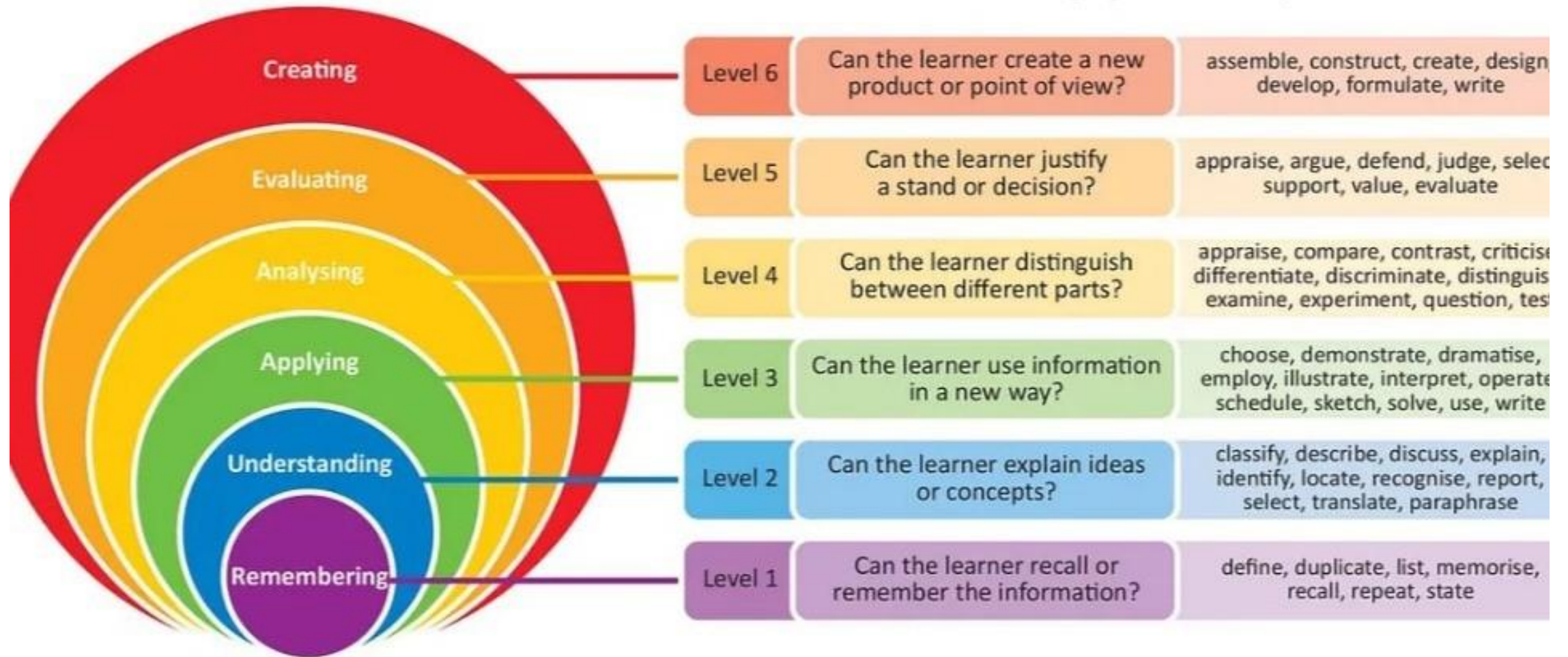
CO-1 ; SO-7; BL-2	b.	Describe the parameters on which organization structure depends	[04]
Q2 CO-1; SO-4; BL-4	b	Describe qualifications/skills set required for project manager and its impact on 'Success and failure of project'	[10]
CO-1 ; SO-7; BL-2	b.	Describe Pure Product (projectized) organization	[04]
Q2 CO-1; SO-4; BL-4	b	"Selection of project manager is critical , important, challenging task for top management" Explain also list the parameters used for screening candidates	[10]

CO-1 ; SO-5 ; BL-2	b.	Describe advantages of Matrix organization structure when one studies project management	[04]
Q2 CO-1; SO-4; BL-4	b	Define 'Project Manager' also Analyze comprehensively the importance and significance of emotional intelligence of project manager on success and outcomes of project	[10]
Q7 CO-1; SO-5; BL-4	b	Compare Traditional(classical) organization structure with Projectized (pure product) organization structure	[04]

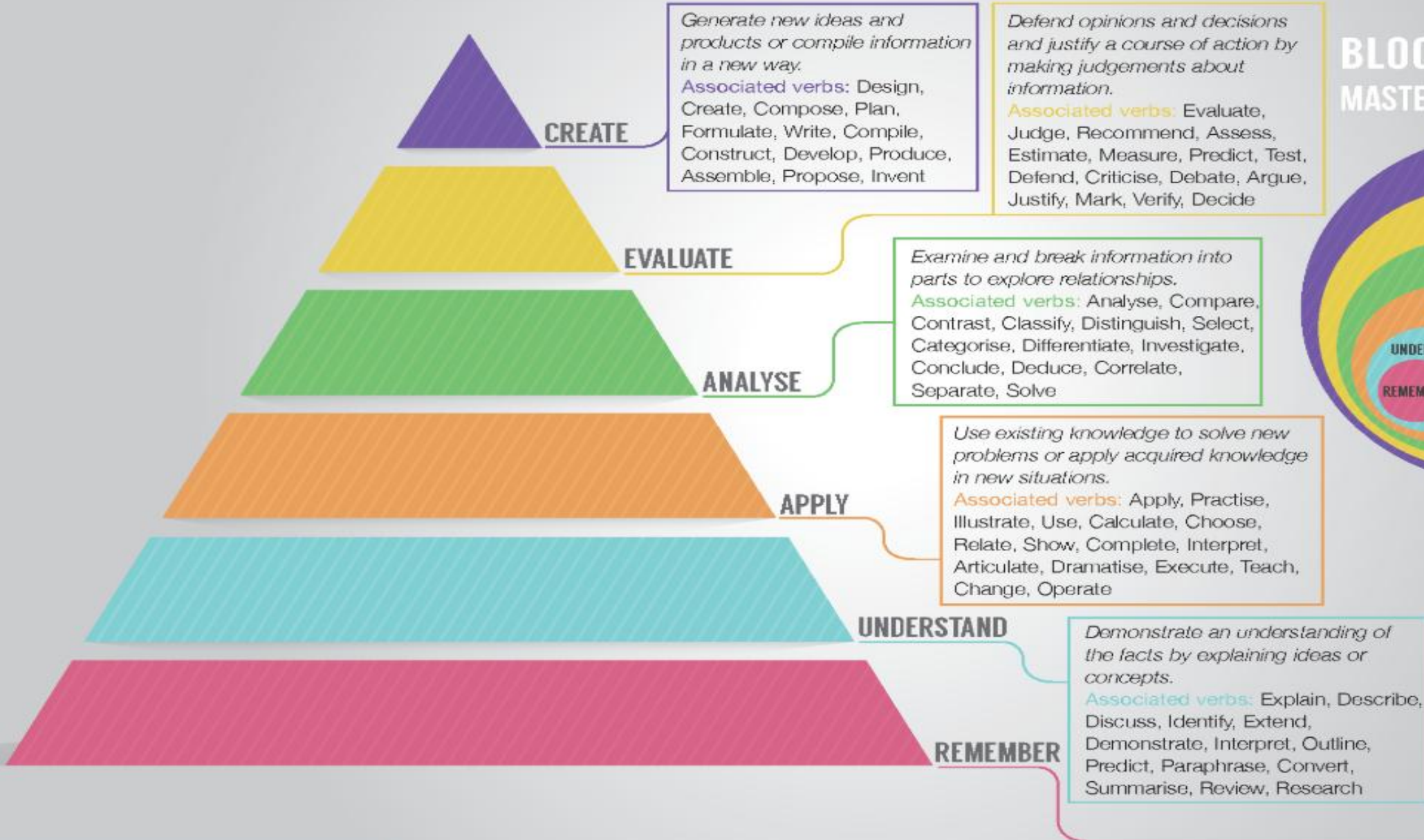
Additional Questions:

1. “Selection of project manager is critical , important, challenging task for top management”
Explain
 2. Describe the selection process of Project Manager
 3. List and explain the expectations from core team members of project
 4. List the requirements the project manager should look while selecting contracted team members for project
 5. Discuss the problems that may arise due to contracted team members
 6. Write detail note on project office
 7. “Integration in large scale projects or mega projects is important and significant as it affects directly and indirectly the important dimensions of success of mega projects” Explain
8. Explain importance & significance of emotional intelligence of project manager on success of project
 9. Why for big/mega projects experience of project manager is more important/valued compare to educational qualification?
 10. Explain how project team affects success as well as failure of project

Bloom's taxonomy (revised)



BLOOM'S REVISED TAXONOMY: RELATED VERBS



BLOOM'S TAXONOMY: MASTERY LEARNING

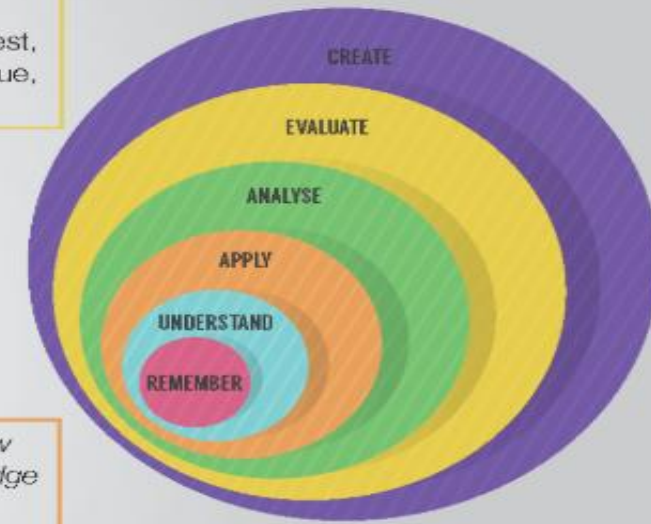


Exhibit memory of previously learned material by recalling facts, concepts and answers.

Associated verbs: Define, Recall, Recognise, Name, Quote, Label, List, Cite, Locate, Find, Reproduce